

PROJECT DOCUMENTATION

PROGRAM DEVELOPMENT PHASES

EduGenie – Google Gemini Powered Learning Assistant

1. Phase 1 – Requirement Understanding

Identify the educational problem and the need for question answering, concept explanation, quizzes, summaries, and learning guidance.

2. Phase 2 – Technology Selection

Select Python and FastAPI for application logic and request handling, Google Gemini for generative AI, Jinja2 for dynamic HTML rendering, and HTML/CSS for the interface.

3. Phase 3 – Backend Development

Create the FastAPI application, configure static files and templates, load the Gemini API key from .env, accept the question using POST form data, send the question to Gemini, and return the generated answer.

4. Phase 4 – Frontend Development

Create the EduGenie webpage with navigation, question input, Ask AI button, dynamic answer section, and feature cards for explanation, quiz, learning path, and summarization.

5. Phase 5 – Integration

Connect the frontend form to the /ask endpoint and connect the backend to Google Gemini through the google-genai SDK.

6. Phase 6 – Testing

Test application loading, question submission, empty input validation, CSS loading, Gemini connection, and server startup.

7. Phase 7 – Demonstration

Present the completed interface and documented output screens for the EduGenie modules.

Prepared for project documentation and academic demonstration.